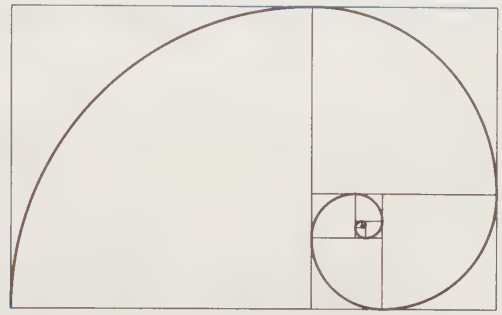
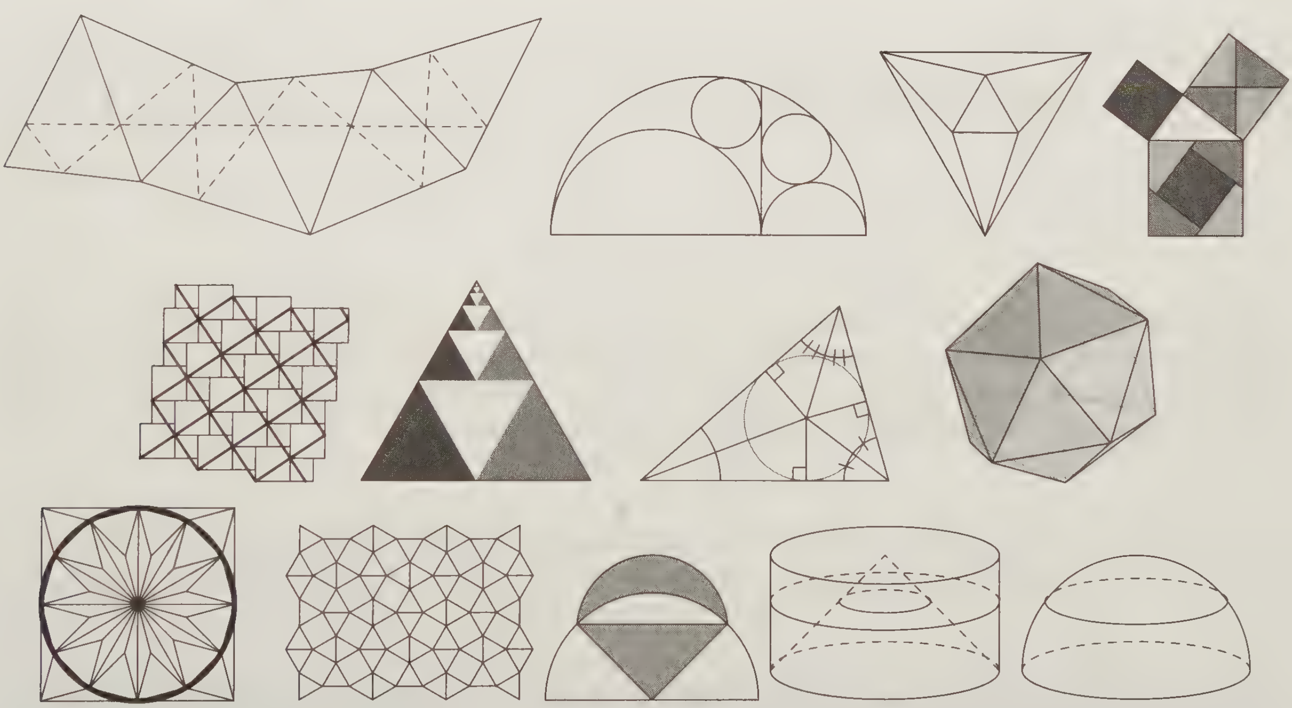


The Golden Ratio Spiral

Do not laugh at notations; invent them, they are powerful. In fact, mathematics is, to a large extent, invention of better notations. – Richard Feynman

CHAPTER 1

What's in a Name?



Each of these images helps tell a story. Throughout this book we'll share these stories with you, but before we tell these stories, we have to name our characters.

1.1 Why Names and Symbols?

To convince you that names and symbols are useful, we'll start at the end of the book instead of the beginning. Here's the final example problem in this book, written without any special symbols or names.

> Draw three points and connect each to the other two with straight paths. Also, draw the circle that passes through all three of these points. Then, draw a line through one of those three points such that the line goes inside the region you just formed and is equally close to the two other straight paths you formed initially through this point. Draw the circle that goes through the one of your three first points you just drew a line through, through the point where this line hits the straight path that connects the other two of your first three points, and through the point that is half-way between these two other points.
>
> Consider the two paths from the point we drew the extra line through to the other two of our first three points. These paths hit our second circle before they hit these other two points. Show that the distance from where the circle hits these paths to the points where these paths end is the same for both paths.

If you can make much sense of this problem, you're a much more careful reader than I am! We need some special names and symbols so we can communicate mathematical ideas more simply.

1.2 Points, Lines, and Planes

Figure 1.1: A Point

A dot. A speck. In geometry, it's a **point**. If you lived on a point, you'd be awfully bored. There would be no up and down, no right and left. You couldn't move any amount in any direction. Since you can't move on your point in any direction, we say a point has 0 **dimensions**. In order to tell one point from another, we usually label them with capital letters, such as point P above.

Figure 1.2: A Segment

Now, say you got so bored on one point that you just had to go to another point. If there were a straight path from one point to another, that path would be called a **line segment**, or just a **segment**. The two points at the ends of a segment are cleverly called the **endpoints** of the segment. We use these endpoints to label the segment. For example, $\overline{AB}$ is the segment from A to B. To denote the length of the segment, we omit the bar. For example, AB equals 1.5 inches in Figure 1.2.

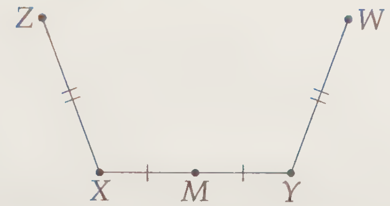


Figure 1.3: A Midpoint and Marking Segments of Equal Length

The endpoints aren't the only points on a segment. There are infinitely many points, since between any two points on the segment, we can find another point. One special point on a segment is the segment's **midpoint**, which is the point halfway between the endpoints. Because the midpoint is the same distance from both endpoints, we say it is **equidistant** from the endpoints. In Figure 1.3, M is the midpoint of $\overline{XY}$. We show that $XM = MY$ in the diagram with the little tick marks along $\overline{XM}$ and $\overline{MY}$. If we have multiple sets of equally long segments, we use a different number of tick marks for each. For example, our diagram above indicates that $ZX = WY$, and that these lengths need not be the same as XM and MY.

Figure 1.4: A Ray

If you're not happy just going from A to B, you can keep going past point B. If you keep going forever, you will make a **ray**. We refer to the ray in Figure 1.4 as $\overrightarrow{AB}$, where the starting point, or **origin**, of the ray comes first. In the diagram, the little arrow indicates that the ray continues forever in that direction.

Figure 1.5: A Line

As you might guess, we could continue forever in both directions. The result is a **line**. Line $\overleftrightarrow{AB}$ is shown in Figure 1.5. We sometimes use a lowercase letter to describe a line, such as line k in the figure. We often leave off the little arrows in the diagrams.

Extra! *[The universe] cannot be read until we have learnt the language and become familiar with the characters in which it is written. It is written in mathematical language, and the letters are triangles, circles and other geometrical figures, without which means it is humanly impossible to comprehend a single word.*

–Galileo Galilei

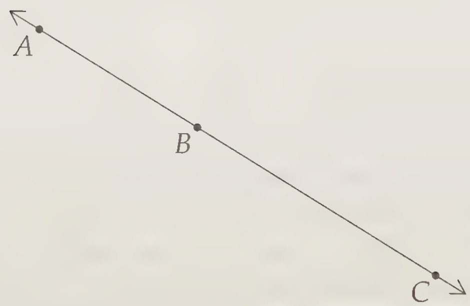


Figure 1.6: Three Collinear Points

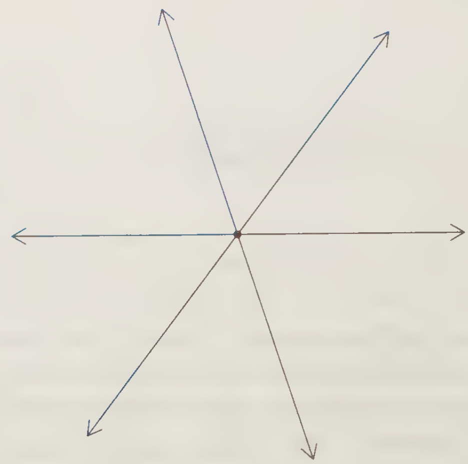


Figure 1.7: Three Concurrent Lines

If three or more points are all on the same line, we say the points are **collinear**, and if three or more lines all pass through the same point, we say the lines are **concurrent**.

Segments, rays, and lines are all one-dimensional figures, since you have only one way you can move along them. Roughly speaking, any path you can draw with a pencil is one-dimensional, meaning you can either move 'forward' on the path or 'backward' on the path.

Once we have the freedom to go off our path and move around on a surface, we're up to two dimensions. On a surface like this page, we might call our dimensions left-right and up-down. If the page extended forever in every direction, we'd call it a **plane**. Most of this book discusses **planar** figures, which are figures that exist in planes.

However, in Chapter 14, we wander off the page and add a third dimension you might think of as 'above-below.' The physical **space** we live in is effectively three-dimensional, and most of what we experience is three-dimensional.

Although it's much harder to think about, there's a great deal of math in higher dimensions. But that's a story for another day.

Exercises

1.2.1 Alice is thinking of a line. How many points on that line does she need to show Bob in order for Bob to know exactly which line she is thinking about?

1.2.2 M is the midpoint of $\overline{AB}$ and N is the midpoint of $\overline{BM}$. If $BN = 4$, then what is AB?

1.2.3 P, Q, R, S, and T are on line k such that Q is the midpoint of $\overline{PT}$, R is the midpoint of $\overline{QT}$, and S is the midpoint of $\overline{RT}$. If $PS = 9$, then what is PT?

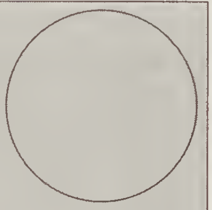
1.2.4★ Points A, B, C, D, and E are five points on a line segment with endpoints A and E. The points are in the order listed above from left to right such that $CD = AB/2$, $BC = CD/2$, $AB = AE/2$, and $AE = 12$. What is the length of $\overline{AD}$? *(Source: MATHCOUNTS)* **Hints:** 203

1.3 Round and Round

Problems

Problem 1.1: Mark a point on a piece of a paper and label it O. Use a ruler to find points on your paper that are 1 inch away from the point O. If you draw all of these points, what figure would you create?

Problem 1.2: The figure shown at right is called a **circle**.



(a) Is it possible to draw a line that does not hit the circle in any points?

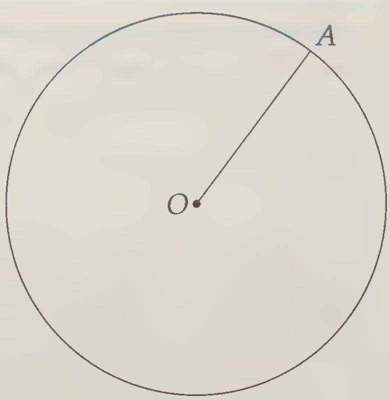
(b) Is it possible to draw a line that hits the circle in exactly one point?

(c) Two points?

(d) Three or more points?

We have many fancy names for things in mathematics. The fancy name we have for a group of points that satisfy certain conditions is a **locus**. While you may never have heard of that word, you've certainly heard of the first locus we'll investigate. (And it's no big deal if you forget the word 'locus' until Intermediate Geometry!)

Problem 1.1: Mark a point on a piece of a paper and label it O. Use a ruler to find points on your paper that are 1 inch away from the point O. If you draw all of these points, what figure would you create?



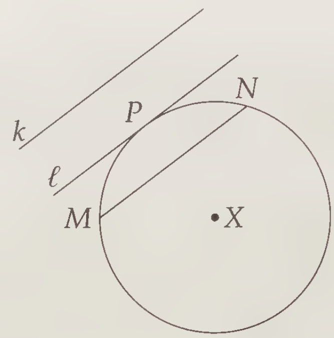
Solution for Problem 1.1: When we draw all the points that are 1 inch away from O, we form a figure called a **circle**. The point O is called the **center** of the circle. We often refer to a circle by its center, writing 'circle O' or '$\odot O$', where the $\odot$ symbol tells us that we're dealing with a circle. We say that $\overline{OA}$ is a **radius** of the circle because it is a segment connecting the center to a point on the circle. We know that all points on the circle must be 1 inch, from the center, so $OA = 1$ inch. The term 'radius' is also used to mean the length of a radius, so we could write: 'The radius of $\odot O$ is 1 inch.'

You'll notice that we didn't use a big dot to mark point A. When there's a label near where two figures meet, the label refers to the point where they meet. Therefore, A is the point where our radius hits the circle. $\square$

Much of our work in this book involves both lines and circles.

Problem 1.2: Can a line and a circle intersect in 0 points? 1 point? 2 points? 3 points? More?

Solution for Problem 1.2: Given $\odot X$, we can clearly find a line that doesn't hit X anywhere. Line k shown below is such a line. Imagine sliding line k closer and closer to $\odot X$ until it touches the circle at exactly one point, such as line ℓ touches $\odot X$ at point P. We say that line ℓ is a **tangent line** to the circle. We can also use 'tangent' as a adjective, and write, 'Line ℓ is tangent to $\odot X$.'

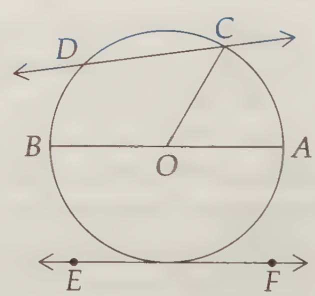


Lines even closer to the center intersect the circle at two points, such as $\overleftrightarrow{MN}$ does. A line that hits a circle at two points is a **secant line**. A segment that connects two points on a circle is a **chord**. $\overline{MN}$ is a chord, while $\overleftrightarrow{MN}$ is a secant line. A chord that passes through the center of a circle is a **diameter**.

Finally, the portion of a circle that connects two points on a circle is called an **arc** of that circle. Of course, we have a symbol for that too: $\widehat{MN}$ is the shorter of the two arcs that connect M and N. We call the shorter of the two arcs that connect two points on a circle a **minor arc** of the circle. The longer arc that connects the two points is a **major arc** of the circle. We usually use three points to denote a major arc: $\widehat{PNM}$ is the longer arc connecting P to M, while $\widehat{PM}$ is the smaller arc connecting them. □

Exercises

1.3.1 In the figure at right, identify whether each of the following is a secant line, a chord, a radius, a diameter, or a tangent line of $\odot O$. (If multiple terms are accurate, list all of the accurate terms.)



(a) $\overline{CO}$

(b) $\overleftrightarrow{EF}$

(c) $\overline{CD}$

(d) $\overline{AB}$

(e) $\overleftrightarrow{CD}$

1.3.2 Suppose point P is outside a given circle. Is it always possible to draw a line through P that is tangent to the circle? (No proof is necessary now; you'll have the tools to prove your answer later in the text.)

1.3.3 What is the maximum number of possible points of intersection of a circle and a triangle? *(Source: AMC 10)* (A triangle is formed by connecting three points with line segments.)

1.3.4★ Two circles and three straight lines lie in the same plane. If neither the circles nor the lines are coincident (meaning the two circles are different and the three lines are all different lines), what is the maximum possible number of points at which at least two of the five figures intersect? *(Source: MATHCOUNTS)* **Hints:** 374

1.4 Construction: Copy a Segment

Classical construction problems are sprinkled throughout the book because a deep understanding of constructions usually leads to a deep understanding of geometry. Construction problems ask us to create precise geometric diagrams with two simple tools. These tools are a **compass**, to make circles, and a **straightedge**, to make straight line segments. Notice that we don't say 'ruler' to make line segments. You don't get to use your straightedge to measure lengths of segments – you can only draw lines. Similarly, you aren't allowed to use your protractor to measure or create angles.

So, what can you do?

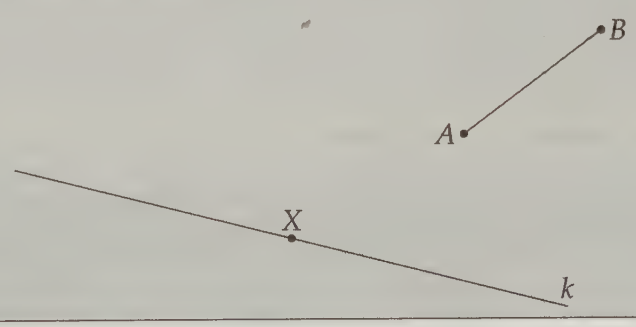
That's the goal of these construction sections: to start learning what you can do with only compass and straightedge. The only operations you can perform with your compass and straightedge are the following:

1. Given a point, you can draw any line through the point.
2. Given two points, you can draw the line that passes through them both.
3. Given a point, you can draw any circle centered at that point.
4. Given a point and a segment, you can draw the circle with its center at that point and with radius equal in length to the length of the segment.
5. Given two points, you can draw the circle through one point such that the other point is the center of the circle.

That's not much, but with these simple operations we can construct an enormous range of diagrams.

Problems

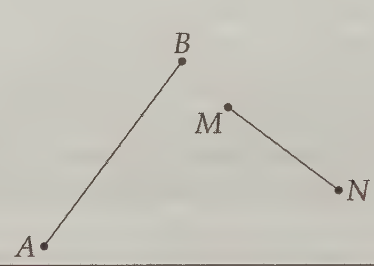
Problem 1.3: Use your compass to find a point Y on k such that $AB = XY$. **You cannot simply use a ruler to measure $\overline{AB}$, then use that measurement to find Y!**



Problem 1.4: Shown below are segments $\overline{AB}$ and $\overline{MN}$.

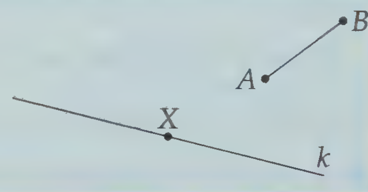
(a) Use straightedge and compass to construct a line segment that has length $AB + MN$. (Reminder: You can't just measure with a ruler!!)

(b) Construct a line segment that has length $AB - MN$. (Even though we don't say 'with a straightedge and compass,' you still can't measure with a ruler! 'Construct' implies 'straightedge and compass' construction.)

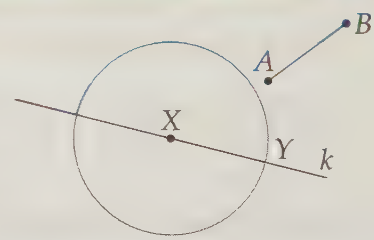


We start our exploration of construction by learning how to copy a segment.

Problem 1.3: Use your compass to find a point Y on k such that $AB = XY$. **You cannot simply use a ruler to measure $\overline{AB}$, then use that measurement to find Y!**



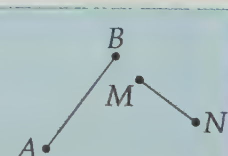
Solution for Problem 1.3: All we can do with a compass is draw circles or parts of circles. To find a point that is AB from X, we first open our compass to a width of AB by putting the point of the compass at A and the compass pencil at B (or vice versa). Then we make a circle with center X and this opening as the radius. Since this $\odot X$ has a radius equal to AB, the two points where it hits k are AB away from X. We can take either one of these as our point Y. □



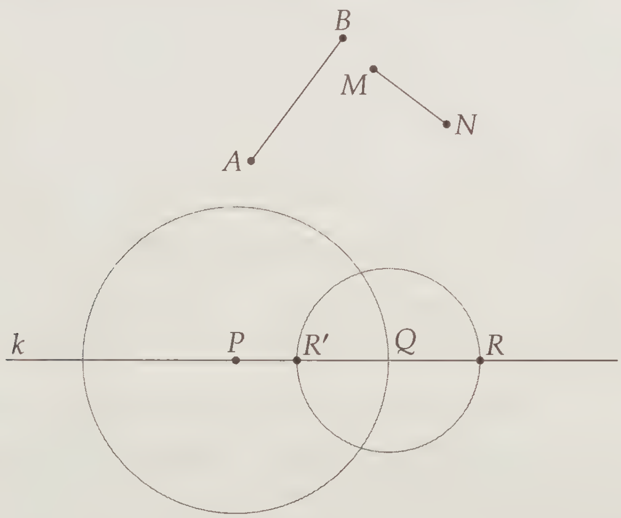
Concept: In nearly all construction problems in which we must make a point, we find that point by constructing two figures that the point must be on. The point we seek is then at the intersection of these two figures. For example, in Problem 1.3, we have line k and construct $\odot X$ that Y must be on. Their intersection gives us the point Y we seek.

Let's try a slightly more challenging construction.

Problem 1.4: Shown are segments $\overline{AB}$ and $\overline{MN}$. Use straightedge and compass to construct a line segment that has length $AB + MN$, and a line segment that has length $AB - MN$.

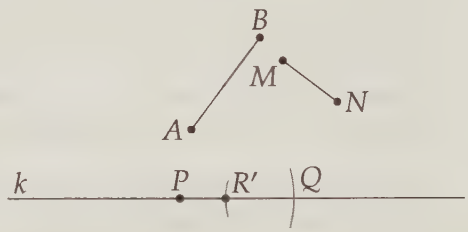


Solution for Problem 1.4: We start by drawing a line k and choosing a point P on line k. We find a point that is $AB+MN$ from P in two steps. First, we find a point a distance of AB from P using our construction technique in Problem 1.3. We do so by opening our compass to width AB and using this radius to draw a circle centered at P. We take one of the points where this circle hits k to be point Q.



Then we find a point that is MN from Q by opening our compass to width MN and using this radius to draw a circle centered at Q. As shown, this circle hits m at two points, R and R'. To get to point R from P, we go a distance of AB to get to Q, then MN more to reach R. Therefore, $PR = AB + MN$. Similarly, to get to R' from P, we first go a distance equal to AB to get to Q, then head *back towards* P a distance of MN to get to R'. So, $PR' = AB - MN$. □

You might have noticed that we didn't need the entire circles we drew in our constructions. We only needed enough of the circle to tell where the circle would hit the line. Typically, these little arcs are all we draw in our constructions. Therefore, our paper when constructing $AB - MN$ in Problem 1.4 might look as shown at right.



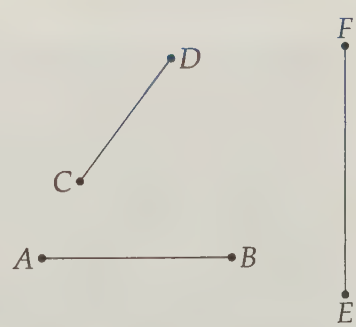
Exercises

1.4.1 Given the segments shown, construct segments with the following lengths:

(a) $AB + CD - EF$.

(b) $2AB$.

(c) $AB - 2EF + 3CD$.



1.5 The Burden of Proof

Earlier we defined a line segment as the direct path that connects two points. It seems obvious that any two points can be connected by a segment. In fact, it seems so obvious that it should be easy to prove. However, it isn't just hard to prove – it's impossible. The statement that any two points can be connected by a straight line segment must be simply accepted as a fact. We call such a statement that must be regarded as fact without proof an **axiom**. Axioms are also sometimes called **postulates**.

When the world's most famous geometer, Euclid, wrote his famous *Elements*, he stated five axioms:

1. Any two points can be connected by a straight line segment.
2. Any line segment can be extended forever in both directions, forming a line.
3. Given any line segment, we can draw a circle with the segment as a radius and one of the segment's endpoints as center.
4. All right angles are congruent. *(We'll talk about right angles and what we mean by 'congruent' shortly!)*
5. Given any straight line and a point not on the line, there is exactly one straight line that passes through the point and never meets the first line.

In Euclid's *Elements*, he combined these axioms to prove ever more complicated mathematical statements. We call such proven mathematical statements **theorems**. A mathematical statement that is not an axiom but hasn't been proved false or true is called a **conjecture**.

In this book, we don't start from Euclid's axioms and prove everything that follows step-by-step. It's a good thing, too! It turns out that even Euclid missed a few axioms. Mathematicians since have shown that Euclid's arguments, in order to be completely valid, would need many more axioms added to these five. In other words, there are some things that even the great Euclid didn't realize are so 'obvious' that they could not be proved. Often when we reach these items in this text, we give a 'common sense' explanation of why we accept these statements as facts. We note when these really are axioms, as opposed to statements that we can prove using previous axioms or theorems.

You can use the proofs we present both as guides for writing your own proofs and as stepping stones to prove interesting theorems of your own.

1.6 Summary

Definitions:

- A **point** is, well, a point. Euclid called a point 'that which has no part.' We can't do much better than that vague description. We typically denote points with capital letters.
- A straight path connecting two points is called a **segment**, and our original two points are the **endpoints** of the segment. We refer to a segment by its endpoints, such as $\overline{AB}$. We remove the bar to denote the length of the segment: AB.

Definitions:

- The point on a segment that is halfway between the endpoints is the **midpoint** of the segment. We also say that this point is **equidistant** from the endpoints.
- If we start at a point, then head in one direction forever, we form a **ray**. Our starting point is the **vertex** of the ray, and we denote a ray as $\overrightarrow{AB}$, where the first point is the vertex of the ray.
- If we continue a line segment past its endpoints forever in both directions, we form a **line**, which we write as $\overleftrightarrow{AB}$.
- If this page were continued forever in every direction, the result would be a **plane**. Since we can move in two general directions, such as right-left and up-down, on a plane, we say the plane has two **dimensions**.
- If we add a third dimension, we are in three-dimensional **space**.

The set of all points that satisfy specific conditions is called a **locus**.

Definitions:

- The set of all points that are the same distance from a given point is a **circle**. The given point is the **center** of the circle, and the fixed distance is the **radius**. We often refer to a circle by its center using the symbol $\odot$, so $\odot O$ refers to a circle centered at O.
- A line that touches a circle at a single point is **tangent** to the circle, while a line that hits a circle at two points is a **secant line**. A segment connecting two points on a circle is a **chord**, and a chord that passes through the center of its circle is a **diameter**. The portion of a circle that connects two points on the circle is an **arc**, which we denote with the endpoints of the arc: $\widehat{MN}$ is the shorter arc that connects M and N.

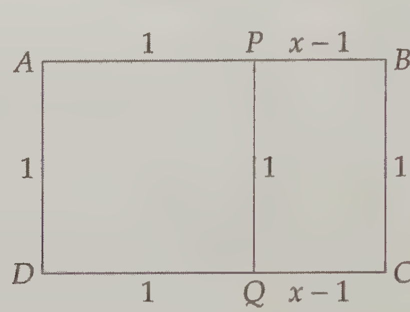
When performing constructions with a straightedge and compass, you can only draw line segments and circular arcs. *You cannot use a ruler to measure segments.* The operations you can perform are:

1. Given a point, you can draw any line through the point.
2. Given two points, you can draw the line that passes through them both.
3. Given a point, you can draw any circle centered at that point.
4. Given a point and a segment, you can draw the circle with its center at that point and with radius equal in length to the length of the segment.
5. Given two points, you can draw the circle through one point such that the other point is the center of the circle.

Extra! *Logic is the art of going wrong with confidence.*

–Morris Kline

Extra! At the top of the first page of each chapter in this book is an image illustrating an interesting geometric fact. The image at the start of this chapter is of the **Golden Ratio Spiral**. A Golden Ratio Spiral is inside a **golden rectangle**, which is a rectangle that can be divided into a square and another rectangle such that the ratio between the dimensions of the new rectangle equals that of the original rectangle.



Shown above is golden rectangle $ABCD$ with dimensions 1 and x. $\overline{PQ}$ divides the rectangle into a square of side 1 and a rectangle with dimensions $x-1$ and 1. Since the ratio of the dimensions of $ABCD$ equals the ratio of the dimensions of $BCQP$, we have

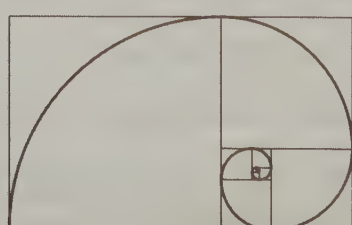
$$\frac{1}{x} = \frac{x-1}{1}.$$

The positive value of x that satisfies this equation is

$$x = \frac{1+\sqrt{5}}{2} \approx 1.618034.$$

This number is the **golden ratio** (also sometimes called the **golden mean**), and is often referred to by the Greek letter ϕ ('phi').

When we divide a golden rectangle into a square and a rectangle, the ratio of the dimensions of the smaller rectangle is the same as that of the original rectangle. Therefore, the smaller rectangle is a golden rectangle too, so we can split it into a square and another smaller golden rectangle. We can do this over and over indefinitely, forming the figure shown below.



All of the squares in the diagram together make up our largest golden rectangle. When we omit the largest square, we get our next golden rectangle. Then we omit the next largest square to find the next golden rectangle, and so on. If we then draw a quarter-circle in each of the squares, as shown above, we get the Golden Ratio Spiral.